test-cube_root_7

October 11, 2026

[25]:
```
prec = 3400
CC = ComplexBallField(prec)
RBF = RealBallField(prec)
Pi = CC(pi, 0)
load('BCG_cubic_git.sage')
load ('cubic_field_K_cube_root_7.sage')
```

```
Loaded 'cubic_field_K_disc_-588.sage'. Contains the following variables:

K: Number Field in y with defining polynomial y^3 - y^2 + 5*y + 1
K has discriminant  -588 = -1 * 2^2 * 3 * 7^2
K has class number  3

pol_K = defining polynomial =  y^3 - y^2 + 5*y + 1
K_basis_SAGE = basis of K =  [1, y, y^2]

yRR = unique real root of pol_K  =  -0.191282440060928 …
yCC = unique complex root of pol_K with positive imaginary part
    =  0.595641220030464 … + I *  2.20750604731611 …
sigmaC = complex embedding of K sending y to yCC
K_basis_SAGE_sigmaC = [1, yCC, yCC^2]
K_basis_SAGE_sigmaR = [1, yRR, yRR^2]

ff =  Fractional ideal (3, y + 1)  of norm  3
q_ff = smallest positive integer in ff = 3
The narrow ray class group of conductor ff has order 6.

aa =  Fractional ideal (5, y - 2)
N_aa = norm of aa =  5
aa_comp = N_aa * aa^(-1)
bb =  Fractional ideal (2, y + 1)  of norm  2
(bb generates the narrow ray class group modulo ff.)

vareps = generator of totally positive units in 1+ff =  -y
varepsRR = real image of vareps =  0.191282440060928
varepsCC = image of vareps under sigmaC
or_Matrix = Matrix with entries the real and complex images of K_basis_SAGE
Eps =  matrix of the fundamental unit vareps in the basis K_basis_SAGE =
```

```
[ 0 -1  0]
[ 0  0 -1]
[ 1  5 -1]

N(aa) = 5 is an odd prime, as (2.9) asks.
aa is coprime to ff, as (2.9) asks.
sigmaR(vareps) = 0.19128 lies in (0,1), as section 2.2 asks.
vareps lies in 1 + ff, as section 2.2 asks.
q_ff = 3 is a positive integer of ff, as section 2.2 asks.

Loaded 'cubic_field_K_cube_root_7.sage'. Contains the following variables:

K: Number Field in y with defining polynomial y^3 - 7
K has discriminant  -1323 = -1 * 3^3 * 7^2
K has class number  3

pol_K = defining polynomial =  y^3 - 7
K_basis_SAGE = basis of K =  [1, y, y^2]

yRR = unique real root of pol_K  =  1.91293118277239 …
yCC = unique complex root of pol_K with NEGATIVE imaginary part
    =  -0.956465591386195 … + I *  -1.65664699997230 …
sigmaC = complex embedding of K sending y to yCC
K_basis_SAGE_sigmaC = [1, yCC, yCC^2]
K_basis_SAGE_sigmaR = [1, yRR, yRR^2]

ff =  Fractional ideal (3, y - 1)  of norm  3
q_ff = smallest positive integer in ff = 3
The narrow ray class group of conductor ff has order 6.

aa =  Fractional ideal (5, y + 2)
N_aa = norm of aa =  5
aa_comp = N_aa * aa^(-1)
bb =  Fractional ideal (2, y + 1)  of norm  2
(bb is a generator of the narrow ray class group modulo ff.)

vareps = generator of totally positive units in 1+ff =  -y + 2
varepsRR = real image of vareps =  0.0870688172276109
varepsCC = image of vareps under sigmaC
or_matrix = Matrix with entries the real and complex images of K_basis_SAGE
Eps =  matrix of the fundamental unit vareps in the basis K_basis_SAGE =
[ 2 -1  0]
[ 0  2 -1]
[-7  0  2]
```

[4]:
```
%%time
L = ff*bb^(-1)
```

```
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
33.102     1     0.030210     76220        (-33/2, 9/2, 3/2)
33.102     1     0.030210     76220        (-27/2, -9/2, -3/2)
82.754     1     0.012084     190549       (-3/2, -21/2, 3/2)
231.71     1     0.0043157    533535       (-63/2, -21/2, -27/2)
264.81     1     0.0037763    609754       (-69/2, -3/2, -21/2)
CPU times: user 1.61 s, sys: 15.4 ms, total: 1.63 s
Wall time: 1.62 s
```

[5]:
```
%%time
N_Ell_Gamma = 77000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-33/2, 9/2, 3/2)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-11/2, 3/2, 1/2)  with adm = (-33/2, 9/2, 3/2)
The wedge for  (-11/2, 3/2, 1/2)  is
a =  (0, 1, -3)
b =  (-1, -2, -5)
alpha = (51/2, -13/2, -5/2) , alpha(b) =  0 alpha(a) =  1
beta = (-13/2, 3/2, 1/2) , beta(a) =  0 beta(b) =  1
gamma = (33/2, -9/2, -3/2)
(alpha,beta,gamma) is positively oriented basis
tau =  1.5347844803 - 0.055221566666*I
sigma =  -0.37524719874 + 0.0048080764950*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [-1/3, 1/30*y + 47/30, -1/30*y^2 - 1/15*y - 1/2]
Wedge negatively oriented with respect to h
certified decimals =  1012
u= 0.030810416506996155 + 0.25196622350817154*I
1/u= 0.47815339872467774 - 3.9103173469562422*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1
```

```
PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 31.7 s, sys: 5.72 ms, total: 31.7 s
Wall time: 31.7 s
```

[6]:
```
%%time
L = ff*bb^(-2)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
16.551     1     0.060420     38110        (33/4, -9/4, -3/4)
16.551     1     0.060420     38110        (27/4, 9/4, 3/4)
16.551     1     0.060420     38110        (87/4, -51/4, 3/4)
115.86     1     0.0086314    266768       (63/4, 21/4, 27/4)
132.41     1     0.0075525    304877       (69/4, 3/4, 21/4)
CPU times: user 1.81 s, sys: 52 ms, total: 1.87 s
Wall time: 1.86 s
```

[7]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (33/4, -9/4, -3/4)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (11/4, -3/4, -1/4)  with adm = (33/4, -9/4, -3/4)
The wedge for  (11/4, -3/4, -1/4)  is
a =  (0, 1, -3)
b =  (-1, -2, -5)
alpha = (9, -2, -1) , alpha(b) =  0 alpha(a) =  1
beta = (7/4, -3/4, -1/4) , beta(a) =  0 beta(b) =  1
gamma = (33/4, -9/4, -3/4)
(alpha,beta,gamma) is positively oriented basis
tau =  1.0695689606 - 0.11044313333*I
sigma =  0.24950560252 + 0.0096161529901*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [1/3, 1/15*y + 17/15, -1/15*y^2 - 2/15*y]
Wedge positively oriented with respect to h
```

```
certified decimals =  1017
u= -0.0022758886902927605 + 0.023672660991712905*I
1/u= -4.0240295453710452 - 41.855951767110186*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1

PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 16.1 s, sys: 12 ms, total: 16.1 s
Wall time: 16.1 s
```

[8]:
```
%%time
L = ff*bb^(-3)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
33.102     2     0.060420     38110        (33/4, -9/4, -3/4)
33.102     2     0.060420     38110        (27/4, 9/4, 3/4)
33.102     2     0.060420     38110        (87/4, -51/4, 3/4)
66.203     1     0.015105     152439       (-69/8, -3/8, -21/8)
165.51     1     0.0060420    381097       (-171/8, 3/8, 21/8)
CPU times: user 1.49 s, sys: 52.6 ms, total: 1.54 s
Wall time: 1.54 s
```

[9]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (33/4, -9/4, -3/4)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (11/4, -3/4, -1/4)  with adm = (33/4, -9/4, -3/4)
The wedge for  (11/4, -3/4, -1/4)  is
a =  (0, 2, -6)
b =  (-2, -4, -10)
alpha = (9, -2, -1) , alpha(b) =  0 alpha(a) =  2
beta = (7/4, -3/4, -1/4) , beta(a) =  0 beta(b) =  2
gamma = (33/4, -9/4, -3/4)
```

```
(alpha,beta,gamma) is positively oriented basis
tau =  1.0695689606 - 0.11044313333*I
sigma =  0.24950560252 + 0.0096161529901*I
The set F has  2 elements. They are
[ 43/8 -11/8  -5/8]
[    0     0     0]
(z, tau, sigma) in K =  [-1/30*y^2 - 1/30*y + 9/10, 1/15*y + 17/15, -1/15*y^2 -
2/15*y]
(z, tau, sigma) in K =  [1/3, 1/15*y + 17/15, -1/15*y^2 - 2/15*y]
Wedge positively oriented with respect to h
certified decimals =  1016
u= -0.83264323668072915 + 3.5502470704680730*I
1/u= -0.062616293827415938 - 0.26698507107383383*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1

PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 29.6 s, sys: 0 ns, total: 29.6 s
Wall time: 29.7 s
```

[24]:
```
%%time
L = ff*bb^(-3)
adm = vector(QQ, [-69/8, -3/8, -21/8])
N_Ell_Gamma = 160000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexFie ld(60)(1/u.mid()))
PK = algdep_K(u, 6, certified_digits(u) - 5)
print('PK=', PK)
```

```
adm= (-69/8, -3/8, -21/8)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-23/8, -1/8, -7/8)  with adm = (-69/8, -3/8, -21/8)
The wedge for  (-23/8, -1/8, -7/8)  is
a =  (4, -1, -13)
b =  (1, -2, -3)
alpha = (149/4, 7/4, 45/4) , alpha(b) =  0 alpha(a) =  1
beta = (-33, -2, -10) , beta(a) =  0 beta(b) =  1
gamma = (69/8, 3/8, 21/8)
(alpha,beta,gamma) is positively oriented basis
tau =  4.2673922401 - 0.027610783333*I
sigma =  -3.6876235994 + 0.0024040382475*I
The set F has  1 elements. They are
[0 0 0]
```

```
(z, tau, sigma) in K =  [-1/3, 1/60*y + 257/60, -1/60*y^2 - 1/30*y - 15/4]
Wedge negatively oriented with respect to h
certified decimals =  1016
u= -0.83264323668072915 + 3.5502470704680730*I
1/u= -0.062616293827415938 - 0.26698507107383383*I
PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 1min 5s, sys: 7.95 ms, total: 1min 5s
Wall time: 1min 5s
```

[26]:
```
%%time
L = ff*bb^(-4)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
33.102     1     0.030210     76220        (69/16, 3/16, 21/16)
82.754     1     0.012084     190549       (171/16, -3/16, -21/16)
231.71     1     0.0043157    533535       (-441/16, 273/16, -9/16)
314.46     1     0.0031800    724083       (363/16, -99/16, 27/16)
314.46     1     0.0031800    724083       (201/16, 207/16, 9/16)
CPU times: user 1.5 s, sys: 28.1 ms, total: 1.53 s
Wall time: 1.53 s
```

[14]:
```
%%time
N_Ell_Gamma = 77000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (69/16, 3/16, 21/16)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (23/16, 1/16, 7/16)  with adm = (69/16, 3/16, 21/16)
The wedge for  (23/16, 1/16, 7/16)  is
a =  (4, -1, -13)
b =  (1, -2, -3)
alpha = (-25/16, 1/16, -9/16) , alpha(b) =  0 alpha(a) =  1
beta = (3/2, -1/2, 1/2) , beta(a) =  0 beta(b) =  1
gamma = (69/16, 3/16, 21/16)
(alpha,beta,gamma) is positively oriented basis
tau =  -0.46521551971 - 0.055221566666*I
sigma =  0.62475280126 + 0.0048080764950*I
```

```
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [1/3, 1/30*y - 13/30, -1/30*y^2 - 1/15*y + 1/2]
Wedge positively oriented with respect to h
certified decimals =  1011
u= 0.47815339872467774 - 3.9103173469562422*I
1/u= 0.030810416506996155 + 0.25196622350817154*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1

PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 31.5 s, sys: 3.99 ms, total: 31.5 s
Wall time: 31.5 s
```

[15]:
```
%%time
L = ff*bb^(-5)
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho          N_Ell_Gamma  adm
16.551     1     0.060420     38110        (-69/32, -3/32, -21/32)
115.86     1     0.0086314    266768       (441/32, -273/32, 9/32)
628.93     2     0.0031800    724083       (201/16, 207/16, 9/16)
827.54     1     0.0012084    1905481      (-1077/32, -339/32, 27/32)
1257.9     1     0.00079500   2896331      (-1587/32, -69/32, -3/32)
CPU times: user 1.8 s, sys: 29.9 ms, total: 1.83 s
Wall time: 1.83 s
```

[17]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-69/32, -3/32, -21/32)
Did not have to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-23/32, -1/32, -7/32)  with adm = (-69/32, -3/32,
-21/32)
The wedge for  (-23/32, -1/32, -7/32)  is
```

```
a =  (4, -1, -13)
b =  (1, -2, -3)
alpha = (19/32, 5/32, 3/32) , alpha(b) =  0 alpha(a) =  1
beta = (-45/16, -11/16, -13/16) , beta(a) =  0 beta(b) =  1
gamma = (69/32, 3/32, 21/32)
(alpha,beta,gamma) is positively oriented basis
tau =  0.069568960574 - 0.11044313333*I
sigma =  -0.75049439748 + 0.0096161529901*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [-1/3, 1/15*y + 2/15, -1/15*y^2 - 2/15*y - 1]
Wedge negatively oriented with respect to h
certified decimals =  1017
u= -4.0240295453710452 - 41.855951767110186*I
1/u= -0.0022758886902927605 + 0.023672660991712905*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1

PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 16 s, sys: 0 ns, total: 16 s
Wall time: 16 s
```

[18]:
```
%%time
L = ff
adm = scan_admissible(10, L, 3)
```

```
the five best admissible vectors, with the number of terms they need for 1000
decimals:
s/rho      s     rho         N_Ell_Gamma  adm
33.102     2     0.060420    38110        (-3, -6, 3)
33.102     2     0.060420    38110        (-12, -9, -3)
33.102     2     0.060420    38110        (-27, -9, 12)
66.203     1     0.015105    152439       (33, -9, -3)
66.203     1     0.015105    152439       (27, 9, 3)
CPU times: user 1.5 s, sys: 12 ms, total: 1.51 s
Wall time: 1.51 s
```

[19]:
```
%%time
N_Ell_Gamma = 39000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
P = algdep_Q(u, 18, certified_digits(u) - 5)
PK=algdep_K(u, 6, certified_digits(u) - 5)
```

```
print('P=',P)
print()
print('PK=',PK)
```

```
adm= (-3, -6, 3)
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (-1, -2, 1)  with adm = (-3, -6, 3)
The wedge for  (-1, -2, 1)  is
a =  (-1, 1, 1)
b =  (-1, -1, -3)
alpha = (-4, -5, 3) , alpha(b) =  0 alpha(a) =  2
beta = (0, 1, -1) , beta(a) =  0 beta(b) =  2
gamma = (3, 6, -3)
(alpha,beta,gamma) is positively oriented basis
tau =  -0.93043103943 - 0.11044313333*I
sigma =  0.24950560252 + 0.0096161529901*I
The set F has  2 elements. They are
[-2 -2  1]
[ 0  0  0]
(z, tau, sigma) in K =  [-1/30*y^2 - 1/30*y - 23/30, 1/15*y - 13/15, -1/15*y^2 -
2/15*y]
(z, tau, sigma) in K =  [-1/3, 1/15*y - 13/15, -1/15*y^2 - 2/15*y]
Wedge negatively oriented with respect to h
certified decimals =  1016
u= -0.062616293827415938 - 0.26698507107383383*I
1/u= -0.83264323668072915 + 3.5502470704680730*I
P= x^18 + 6*x^17 + 1782*x^16 - 3441*x^15 + 53490*x^14 - 111702*x^13 +
544386*x^12 - 1162674*x^11 + 1975422*x^10 - 2172665*x^9 + 1975422*x^8 -
1162674*x^7 + 544386*x^6 - 111702*x^5 + 53490*x^4 - 3441*x^3 + 1782*x^2 + 6*x +
1

PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 29.4 s, sys: 2  s, total: 29.4 s
Wall time: 29.4 s
```

[20]:
```
%%time
L = ff
adm = vector(QQ, [27, 9, 3])
N_Ell_Gamma = 160000
print('adm=', adm)
u = EllGammaunit_aa_h(adm/q_ff, L)
print('u=', ComplexField(60)(u.mid()))
print('1/u=', ComplexField(60)(1/u.mid()))
PK = algdep_K(u, 6, certified_digits(u) - 5)
print('PK=', PK)
```

```
adm= (27, 9, 3)
```

```
Had to change signs of a and b to satisfy Lemma 2.1 (ii).
Evaluating at h = adm/3 = (9, 3, 1)  with adm = (27, 9, 3)
The wedge for  (9, 3, 1)  is
a =  (-1, 5, -6)
b =  (0, 1, -3)
alpha = (8, 3, 1) , alpha(b) =  0 alpha(a) =  1
beta = (-22, -8, -3) , beta(a) =  0 beta(b) =  1
gamma = (27, 9, 3)
(alpha,beta,gamma) is positively oriented basis
tau =  0.26739224014 - 0.027610783333*I
sigma =  -0.68762359937 + 0.0024040382475*I
The set F has  1 elements. They are
[0 0 0]
(z, tau, sigma) in K =  [1/3, 1/60*y + 17/60, -1/60*y^2 - 1/30*y - 3/4]
Wedge positively oriented with respect to h
certified decimals =  1016
u= -0.062616293827415938 - 0.26698507107383383*I
1/u= -0.83264323668072915 + 3.5502470704680730*I
PK= x^6 + (6*y^2 - 14*y + 2)*x^5 + (4*y^2 - 6*y + 2)*x^4 + (106*y^2 - 152*y -
103)*x^3 + (4*y^2 - 6*y + 2)*x^2 + (6*y^2 - 14*y + 2)*x + 1
CPU times: user 1min 5s, sys: 3.93 ms, total: 1min 5s
Wall time: 1min 5s
```

[ ]: